

Interaction of mycotoxin zearalenone with human serum albumin



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ABSTRACT

Zearalenone (ZEN) is a mycotoxin produced mainly by *Fusarium* species. Fungal contamination of cereals and plants can result in the formation of ZEN, leading to its presence in different foods, animal feeds, and drinks. Because ZEN is an endocrine disruptor, it causes reproductive disorders in farm animals and hyperoestrogenic syndromes in humans. Despite toxicokinetic properties of ZEN were studied in more species, we have no information regarding the interaction of ZEN with serum albumin. Since albumin commonly plays an important role in the toxicokinetics of different toxins, interaction of ZEN with albumin has of high biological importance. Therefore the interaction of ZEN with human serum albumin (HSA) was investigated using spectroscopic methods, ultrafiltration, and molecular modeling studies. Fluorescence spectroscopic studies demonstrate that ZEN forms complex with HSA. Binding constant (K) of ZEN-HSA complex was quantified with fluorescence quenching technique. The determined binding constant ($\log K = 5.1$) reflects the strong interaction of ZEN with albumin suggesting the potential biological importance of ZEN-HSA complex formation. Based on the results of the investigations with site markers as well as docking studies, ZEN occupies a non-conventional binding site on HSA. Considering the above listed observations, we should keep in mind this interaction if we would like to precisely understand the toxicokinetic behavior of ZEN.

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1. Introduction

The foodborne mycotoxin zearalenone (ZEN) is produced by several *Fusarium* species [1]. ZEN occurs principally in cereals (e.g. maize, barley, rye, wheat, and sorghum), however, other sources of the mycotoxin were described as well, such as nuts, edible oils, spices, milk, beer, and drinking water [1–3]. The wide occurrence and high thermal stability of ZEN (it is stable up to 150 °C) are making very difficult its eradication from the food chain [4]. ZEN is an endocrine disruptor molecule because of its xenoestrogenic effect on animals and humans [5]. The effect of ZEN on estrogen receptors leads to reproductive disorders in farm animals as well as hyperoestrogenic syndromes in humans [1,6]. In addition, further toxic impacts of ZEN are also described in previous studies (e.g. haematotoxic, hepatotoxic, or genotoxic effects) [1,7].

Albumin is the most abundant plasma protein in the blood. Human serum albumin (HSA) maintains the oncotic pressure and the pH in the human circulation as well as one of its very important functions is the complex formation with several endogenous (e.g., bilirubin and

fatty acids) and exogenous (e.g., drugs and xenobiotics) compounds [8]. Thus, HSA commonly plays a crucial role in the pharmacokinetics and toxicokinetics of drugs and toxins, respectively [9,10]. Albumin is built up from three domains (I, II, and III), and each domain contains two subdomains (A and B). There are two main binding site on albumin for most of the drugs and xenobiotics, Sudlow's Site I (located on subdomain IIA) and Sudlow's Site II (on subdomain IIIA) [8]. However, recent studies highlighted that a third binding site on subdomain IB (Heme binding site) is also commonly involved [11]. Formation of stable non-covalent complexes of aflatoxin B1, citrinin, deoxynivalenol, and patulin with HSA were described in previous studies [12–15]. Furthermore, highly stable interaction of HSA with ochratoxin A was demonstrated by many studies as well [10]. Despite several toxicokinetic studies performed with ZEN investigating more species (including man) [16–20], we have no information regarding ZEN-albumin interaction.

In this study the interaction of ZEN with human serum albumin was investigated applying fluorescence spectroscopic methods as well as ultrafiltration and molecular modeling studies. Herein, we demonstrate that ZEN forms stable complex with albumin, occupying a non-conventional binding site on HSA. Our data suggest that ZEN-HSA interaction may play an important role in the toxicokinetics of ZEN.

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2. Materials and Methods

2.1. Reagents

All of the applied reagents and solvents were of spectroscopic or analytical grade. Zearalenone (ZEN), human serum albumin (HSA; purity: >96%), bilirubin (BIL), ibuprofen (IBU), L-thyroxine, ochratoxin A (OTA), phenylbutazone (PBZ), and warfarin (WAR) were obtained from Sigma. 5000 μM stock solution of ZEN was prepared in ethanol (Reanal, spectroscopic grade) and stored at -20°C . In order to mimic extracellular physiological conditions, measurements were performed in phosphate buffered saline (PBS, pH 7.4).

2.2. Spectroscopic Measurements

Fluorescence spectroscopic measurements were carried out by Fluorolog $\tau 3$ spectrofluorometric system (Jobin-Yvon/SPEX) and Hitachi F-4500 fluorescence spectrophotometer. UV-VIS spectra were recorded applying Specord Plus 210 (Analytic Jena AG, Jena, Germany) and HALO DB-20 (Dynamica, London, UK) spectrophotometers. All analyses were performed at 25°C .

Fluorescence anisotropy (r) data were determined using the following equation:

$$r = \frac{(I_{VV} - G * I_{VH})}{(I_{VV} + 2 * G * I_{VH})} \quad (1)$$

where I_{VV} and I_{VH} are fluorescence intensities measured in vertical position of polarizer at pre-sample site and at vertical and horizontal position of post-sample polarizer, respectively; while G is the instrumental factor.

Considering the additive behavior of anisotropy, in a mixed system where both ZEN and HSA show fluorescence property as well as assuming absence of interaction between ZEN and HSA, the minimal value of fluorescence anisotropy of the samples can be calculated as the following:

$$r = f_{ZEN} * r_{ZEN} + f_{HSA} * r_{HSA} \quad (2)$$

where r is fluorescence anisotropy of the sample, f_{ZEN} and f_{HSA} are the fractions of ZEN and HSA in the solution, respectively; while r_{ZEN} and r_{HSA} are the anisotropies of ZEN and HSA alone, respectively. If measured fluorescence anisotropy values exceed the values calculated using Eq. (2), that reflects the interaction of ZEN with HSA.

Binding constant of ZEN-HSA complex was calculated applying Hyperquad2006 program package, assuming 1:1 stoichiometry [21,22].

To investigate the role of drug binding site I regarding ZEN-HSA complex formation, interaction of ZEN with warfarin and ochratoxin A was investigated in the presence of HSA, using our previously described models with minor modifications: (1) Increasing amounts of ZEN (0–10 μM) were added to 1 μM warfarin and 3.5 μM HSA (in PBS, pH 7.4) then fluorescence emission spectra were recorded using 317 nm excitation wavelength [13,23]; (2) Increasing amounts of ZEN or warfarin (0–50 μM) were added to 1 μM OTA and 1.4 μM HSA (in PBS, pH 7.4) then fluorescence anisotropy was determined applying 393 and 446 nm excitation and emission wavelengths, respectively [13,24].

2.3. Thermodynamic Studies

The thermodynamic parameters, enthalpy and entropy changes associated to the complex formation between ZEN and HSA molecules were calculated as follows: the logarithms of binding constants ($\log K$) at different temperatures were plotted against the reciprocal temperature according to the van't Hoff equation. The enthalpy change was derived from the slope while the entropy change was derived from the

intercept of the line fitted to the data measured at different temperatures:

$$\log K = \frac{-\Delta H}{2.303 \cdot R} \cdot \frac{1}{T} + \frac{\Delta S}{2.303 \cdot R} \quad (3)$$

where ΔH and ΔS reflect the enthalpy and entropy change of the association reaction, while R is the gas constant and T refers the temperature.

2.4. Ultrafiltration

Ultrafiltration studies were performed using the procedure described earlier [13]. In order to remove glycerol, Amicon Ultra-4 centrifugal filter units (10 kDa molecular weight cut-off; from Merck Millipore) were washed twice with 3 ml water then with 3 ml PBS. Samples (2.5 ml) were transferred into the filter after which centrifuged at 7500g (fixed-angle rotor) for 10 min at 25°C . Thereafter, WAR concentrations were quantified in the filtrate based on the fluorescence intensity of WAR ($\lambda_{\text{exc}} = 309 \text{ nm}$, $\lambda_{\text{em}} = 389 \text{ nm}$) using a calibration curve (0.1–1.0 μM). No spectral interferences were observed during these experiments because both ZEN (possibly because of its adsorption by the filter) and HSA (because of the large size of the protein) are unable to reach the filtrate. Regardless of the concentration used (0.2–1.0 μM WAR concentrations were tested), constant recovery of WAR was observed ($80.3 \pm 2.1\%$).

2.5. Structural Calculations

2.5.1. Ligand Preparation

The zearalenone molecule was built in Maestro [25]. The raw structure was energy minimized, using the semi-empirical quantum chemistry program package, MOPAC [26] and the PM6 parameterization. The gradient norm was set to 0.001. The energy minimized structure was subjected to force calculations. The force constant matrices were positive definite.

2.5.2. Target Preparation

An apo crystallographic structure (PDB code: 1ao6) was used as target a molecule in our calculations. Acetyl and amide capping groups were attached to the N- and C-termini, respectively, using the Schrödinger Maestro program package v. 9.6 [25]. As 1ao6 contains a homodimer structure, only chain A was used for calculations. Co-crystallized ions and water molecules were removed before minimizing the protein structure. The target molecule was minimized using a two-step protocol with the GROMACS [27] software package, including a steepest descent and a conjugate gradient step, using AMBER99-ildn force field [28]. Exit tolerance levels were set to 1000 and 10 $\text{kJmol}^{-1} \text{ nm}^{-1}$ while maximum step sizes were set to 0.5 and 0.05 nm, respectively. His464 was modeled in three forms, with proton at the epsilon, delta, and both positions, respectively.

2.5.3. Blind Docking

Using the optimized ligand and target structures, blind docking calculations were performed with AutoDock 4.2 [29] program package as described in our previous publications [13,30,31]. Gasteiger-Marsilli partial charges were added to both ligand and target atoms, using AutoDock Tools [29] and a united atom representation was applied for non-polar moieties. A grid box of $200 \times 200 \times 200$ points, and 0.375 Å spacing was calculated and centered on the center of mass of the target by AutoGrid 4. Lamarckian genetic algorithm was used for global search. Flexibility at two active torsions was allowed on the ligand, number of docking runs was set to 100, numbers of energy evaluations and generations were 20 million [32]. The docked ligand copies were ordered according to AutoDock 4 scores [33], and subsequently clustered using a tolerance of 1.75 Å root mean square deviation (RMSD) between cluster

members [31]. The cluster representatives with the lowest AutoDock 4 scores [33] were further analyzed.

2.5.4. Post-docking Molecular Dynamics (MD) Simulations

MD simulations of 100 ns length were performed for the top three ranked HSA-bound ligand conformations from the blind docking calculations. For the first rank, additional two MD simulations were performed for investigation of the impact of the above-mentioned His464 protonation states on the stability of the binding mode of the ligand. The five ligand-target complexes (see Results and Discussion) were performed with the GROMACS program package using force field settings described at section “Target Preparation”. The Parrinello-Rahman temperature and pressure coupling method was used in order to sample an NPT environment. Protein and non-protein groups were coupled separately, with a reference temperature 300 K and reference pressure 1 bar. No position restraint was applied on the target or ligand atoms, during simulations. The Particle Mesh Ewald method was used for long-range electrostatics, and the van der Waals and Coulomb cut-offs were set to 11 Å. All 1000 frames generated in MD trajectory were extracted. In order to re-evaluate the strength of target-ligand binding, intermolecular interaction energies were calculated using the Lennard-Jones potential (E_{inter}) with Amber2012 parameters [34].

2.6. Statistics

Statistical analyses were performed by one-way ANOVA test (IBM SPSS Statistics, Version 21). The level of significance was set at $p < 0.05$.

3. Results and Discussion

3.1. Effect of Human Serum Albumin on the Fluorescence of Zearalenone

Two peaks appear in the fluorescence excitation spectrum of ZEN, these peaks show maxima at 275 and at 315 nm [35]. During these experiments, 315 nm was applied as excitation wavelength because of its

larger distance from the excitation maximum of HSA (280 nm). In order to investigate the complex formation of ZEN with HSA, increasing quantities of HSA (0–10 μM) were added to standard amount of ZEN (2 μM). Using 315 nm as excitation wavelength, ZEN shows its fluorescence emission maximum at 455 nm in PBS. Fig. 1A demonstrates emission spectra of ZEN in the absence and in the presence of increasing HSA concentrations. The changes of the fluorescence spectra of ZEN are partly derived directly from the signal of the HSA preparation (Fig. 1B). Since these peaks (possibly resulted from the presence of some impurity in the HSA preparation) gave a maximum at lower wavelength (410 nm) compared to ZEN (455 nm), a blue shift of the combined spectrum can be observed. Furthermore, if we subtract the spectra of Fig. 1B from the spectra of Fig. 1A, we can recognize that the fluorescence maximum of ZEN is unchanged (Fig. S1). In Fig. 1C, we illustrated the fluorescence emission intensities (recorded at 455 nm) of Fig. 1A (I_1), Fig. 1B (I_2), and the differences of these intensities ($I_1 - I_2$). The strong fluorescence enhancement of ZEN ($I_1 - I_2$) in the presence of HSA suggests the formation of ZEN-HSA complex.

To confirm the existence of ZEN-HSA complexes, fluorescence anisotropy studies were performed. Fluorescence anisotropy of samples containing ZEN, HSA and ZEN + HSA were measured using 315 and 455 nm as excitation and emission wavelengths, respectively. Considering the additive behavior of anisotropy in a mixture of non-interacted species, we can calculate the expected anisotropy values assuming absence of any interaction between the two compounds (see Eq. (2)). Fig. 1D demonstrates the measured and the calculated anisotropy data. The measured anisotropy of samples containing both ZEN and HSA strongly exceeds the anisotropy values calculated from the anisotropy of separated ZEN and HSA. This observation reflects the presence of ZEN-HSA complexes in the solution. Because the complex formation of a small molecule like ZEN with a macromolecule (HSA) leads to the considerable decrease of its rotational freedom, fluorescence anisotropy values of HSA-bound ZEN significantly increased. Therefore, it is plausible to conclude that the elevation of fluorescence anisotropy data in the mixed ZEN + HSA system (compared to the calculated values where the

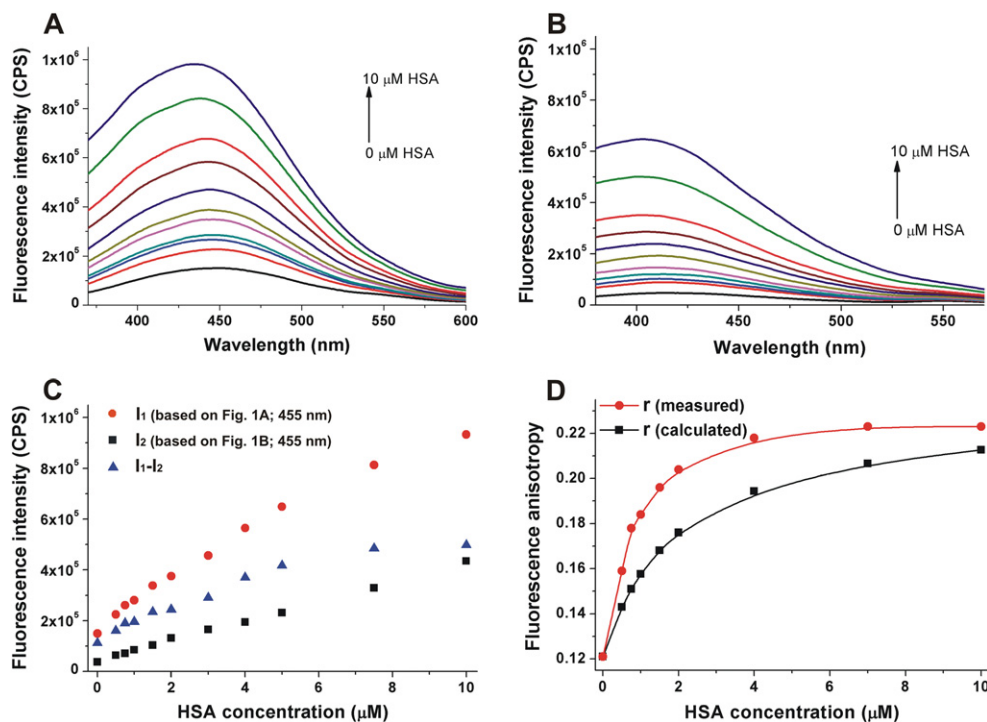


Fig. 1. (A): Fluorescence emission spectra of ZEN (2 μM) in the presence of HSA (0–10 μM). (B): Fluorescence emission spectra in the absence of ZEN (concentrations of HSA are varied within 0–10 μM , as before). (C): Fluorescence emission intensities at 455 nm based on panel A (I_1), panel B (I_2), and the differences of these two intensities ($I_1 - I_2$). (D): Fluorescence anisotropy data of ZEN (2 μM) in the presence of increasing HSA levels (0–10 μM) compared to the calculated data assuming the absence of complex formation (see Eq. (2)) [$\lambda_{\text{exc}} = 315 \text{ nm}$, $\lambda_{\text{em}} = 455 \text{ nm}$; exc slit width: 4 nm, em slit width: 4 nm].

absence of complex formation was assumed) are resulted from the increased anisotropy values of ZEN due to its interaction with HSA.

In the next series of experiments, ZEN were added with increasing concentrations (0–10 μM) to standard amount of HSA (2 μM). During these measurements 315 nm was applied again as excitation wavelength. Fluorescence emission spectra of ZEN were recorded in the absence (Fig. 2A) and in the presence (Fig. 2B) of HSA. Fig. 2C shows the emission intensities of the samples at 455 nm plotted against ZEN concentrations. Fig. 2 highlights that the fluorescence signal of ZEN increases in much stronger extent in the presence of HSA. These observations highlight again that ZEN forms complex with HSA resulting in the fluorescence enhancement of the mycotoxin.

3.2. Fluorescence Quenching Studies

To quantify the stability of ZEN-HSA complex, fluorescence quenching studies were performed. Increasing amounts of ZEN (0–5 μM) were added to standard 2 μM HSA. Using 280 nm as excitation wavelength, HSA shows emission maximum at 340 nm. ZEN is also shows considerable emission ($\lambda_{\text{em, max}} = 455 \text{ nm}$) after excitation at 280 nm. However, the spectral interference can be neglected under the applied conditions, because there is a great difference between the emission maxima of HSA and ZEN. Furthermore, the absorbance value of the highest applied concentration of ZEN (5 μM) was approximately 0.05 at 280 nm in PBS, which cannot cause relevant inner filter effect. Fig. 3 demonstrates that even nanomolar ZEN concentrations led to considerable quenching of the fluorescence signal of HSA. The remarkable decrease of the fluorescence of tryptophan residue even in the presence of very low ZEN concentrations suggests that the binding site of ZEN on HSA is located close to tryptophan-214. Using 340 nm as emission wavelength, the binding constant was quantified by non-linear fitting. Our calculations suggest 1:1 stoichiometry during the complex formation. The determined $\log K$ value was 5.09 ± 0.01 suggesting strong interaction between ZEN and HSA. The magnitude of the complex stability suggests the extensive albumin-binding of ZEN in the human circulation if we consider some examples regarding drug-HSA interactions like warfarin ($\log K = 5.4$, 99% is HSA-bound in the circulation), furosemide ($\log K = 5.2$, 96–99% is HSA-bound in the circulation), or phenytoin ($\log K = 4.3$, 89% is HSA-bound in the circulation) [23,36–38].

3.3. Thermodynamic Studies

The thermodynamic parameters associated to the interaction of ZEN with HSA was determined using the van't Hoff equation (Fig. 4). -30.09 kJ/mol and $-3.5 \text{ J/K}\cdot\text{mol}$ were determined as the enthalpy and entropy changes during the complex formation, respectively. This

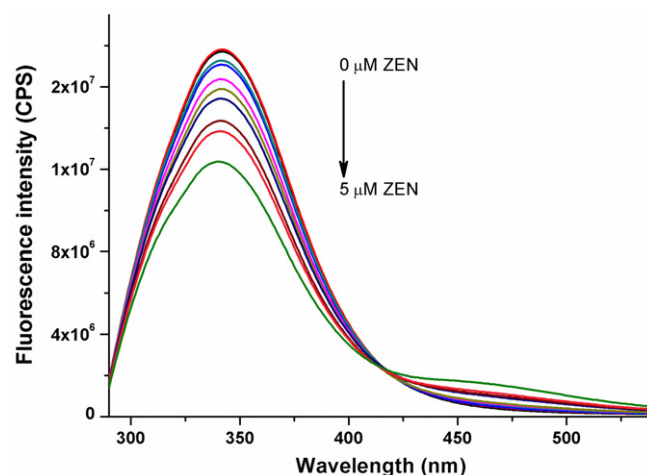


Fig. 3. Fluorescence quenching of HSA (2 μM) by increasing ZEN concentrations (0–5 μM) in PBS [$\lambda_{\text{exc}} = 280 \text{ nm}$; exc slit width: 4 nm, em slit width: 4 nm].

result reflects enthalpy driven interaction between the ZEN and HSA molecules. Comparing this values with the parameters obtained for the citrinin-HSA interaction ($\Delta H = -24.15 \text{ kJ/mol}$ and $\Delta S = 20.90 \text{ J/K}\cdot\text{mol}$) [13], we can conclude that the stronger molecular interaction, i.e. higher enthalpy change is associated with smaller entropy gain in agreement with the enthalpy – entropy compensation. Considering that the stability constant of ZEN-HSA interaction is very close to the citrinin-HSA interaction at room-temperature, the lower entropy gain in the former reaction reflects that the ZEN molecules keep their solvation shell during the interaction process.

3.4. Structural Calculations

To identify the binding site of ZEN on HSA, blind docking simulations [30,31] and post-docking MD calculations with explicit water model were performed. Blind docking resulted in nine representative binding modes of which the top three ranks with the best target-ligand interaction energies (Table 1) were further studied by MD.

Subscripts 1–3 of Site A in Table 1 denote results obtained with epsilon-, delta-, and doubly (both epsilon and delta) protonated His464 side-chain. Site A₁ is located between Sudlow's site I and II, Site B is located in subdomain IA but it is different from the most buried fatty acid site previously described in this subdomain, while Site C is the Heme-Fe binding site (see Fig. 5A) [8]. A relatively small difference could be observed between E_{inter} values at Sites A–C after docking calculations,

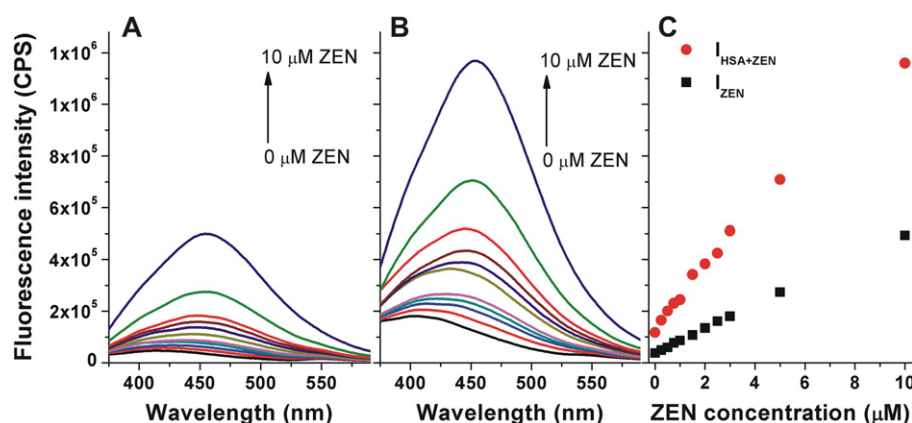


Fig. 2. Fluorescence emission spectra of increasing ZEN concentrations (0–10 μM) in the absence of HSA (A). Fluorescence emission spectra of increasing ZEN concentrations (0–10 μM) in the presence of 2 μM HSA (B). Fluorescence emission intensities of ZEN (I_{ZEN}) and HSA + ZEN ($I_{\text{HSA+ZEN}}$) at 455 nm (C) [$\lambda_{\text{exc}} = 315 \text{ nm}$; exc slit width: 4 nm, em slit width: 4 nm].

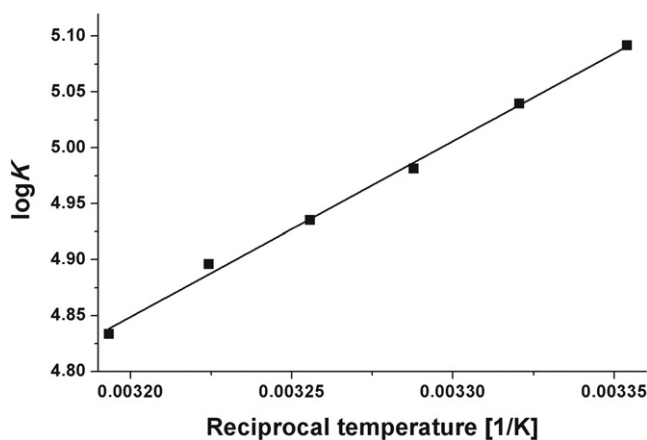


Fig. 4. Van't Hoff plot associated to the interaction of ZEN with HSA in PBS buffer.

indicating that they are all equally good candidates for ZEN binding. In order to further distinguish the most favorable and stable binding site the docked HSA-ZEN complexes were submitted to post-docking MD simulations with explicit water model. Such MD calculations allow refinement of bound ligand positions and also re-ranking of the sites as they account for target flexibility and free thermal motion of explicit water molecules [39,40]. Considering that binding site residue His464 (Site A₁₋₃) can adopt various protonation states, three separate MD simulations were performed for Site A₁ with His464 protonated either on N_{epsilon}, N_{delta} or both nitrogen atoms (Table 1).

After the 100 ns simulations, ZEN remained bound to all binding sites (Table 1), indicating that the docked positions of ZEN were realistic. However, a re-ranking of ZEN positions from docking calculations was observed after MD simulations. Site C (the Heme binding site) got the 1st rank as the most favorable site from interaction energy point of view after MD. This result is not surprising as such a large binding site as Site C can accept various ligands with multiple/strong binding positions if not occupied by heme.

Whereas the above occupancy of Site C by ZEN was somewhat expectable, stable binding of ZEN to Site A is of interest from a functional viewpoint. The binding position that was ranked the 1st after docking, was the most mobile in the binding site, among the investigated positions (see Fig. 5), but did not dissociate, form the site between Sudlow's site I and II (Fig. 5B). From the MD results, it follows, that ZEN is most probably binds to the site A, when is His464 in positively charged protonation state (A₃, Fig. 5B). This results is suggested by the favorable E_{inter} value, compared to the epsilon protonation, but also by the fact that during this simulations, ZEN was more stable in the binding pocket when His464 was protonated on both nitrogen atoms.

After MD simulations, Glu465 remained in contact with the ZEN in all of the three cases of site A. Thr478 remained in contact with the ligand only in cases A₂ and A₃ (best binder cases). At A₃ the interaction with Arg484 and Glu465 are crucial in stabilizing the ligand by two strong

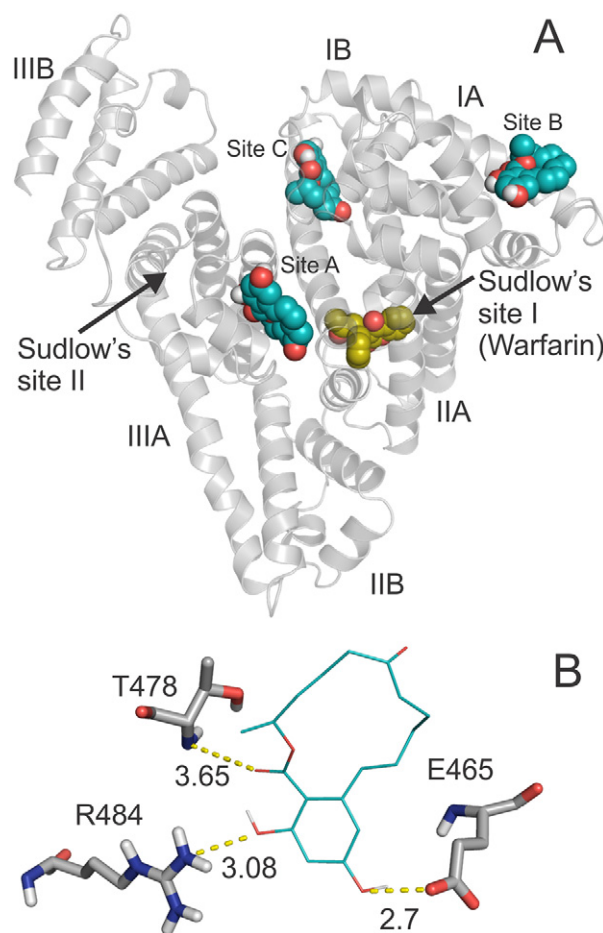


Fig. 5. Top binding sites of ZEN on HSA as found by blind docking and subsequent molecular dynamics calculations (A). The HSA target is in grey cartoon, ZEN positions representing the top three Sites A₃-C, and a reference position of warfarin (adopted from PDB 1h9z, yellow) as bound to Sudlow's site I are in spheres. (B) A close-up of Site A₃ with the representative conformation of ZEN after the MD simulation (thin teal for carbon sticks in the middle) and HSA binding site residues in the surrounding (thick grey for carbon sticks). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

H-bonds. At the same time, the docked ZEN was replaced by water molecules at His464 after MD refinements. This is partly due to the repulsive effect of the charged, highly hydrophilic His464 and the hydrophobic part of ZEN. Thus, His464 prefers interactions with water molecules replacing the ligand which is fairly common in such situations [40].

3.5. Effects of Site Markers on ZEN-HSA Interaction

To get closer insights on the favorite binding site of ZEN on HSA, three site markers were applied, each of them is associated to different

Table 1
Results of blind docking and post-docking MD simulations.

Site ^a	HSA residues ^b	Blind docking		Molecular dynamics	
		Rank	E _{inter} (kcal/mol)	Rank	E _{inter} (kcal/mol)
A ₁	Lys205 , His464, Glu465, Pro468, Thr474, Cys477, Thr478, Arg484, Phe206 , Glu465	1	−20.65	5	−17.58
A ₂	Cys461 , Glu465 , Cys477 , Thr478	–	–	3	−26.28
A ₃	Glu465 , Thr478 , Arg484	–	–	2	−27.77
B	Phe49 , Thr52, Glu60, Asn61, Lys64, Leu69, Asp72, Lys73 , Asp63	2	−20.37	4	−19.94
C	Leu115, Arg117, Tyr138 , Ile142, His146 , Phe149, Leu185, Arg186, Gly189 , Lys190 , Tyr161 , Leu182	3	−17.05	1	−36.76

^a In the case of site A, three protonation states of HSA residue His464, and the corresponding three target structures were investigated in the MD simulations of target-ligand complexes.

^b HSA binding site residues within 3.5 Å from the cluster representative ligand structure. Bold: after MD, bold & italic: after blind docking and MD, normal: after blind docking.

binding sites. Each tested compound binds to HSA with high affinity, occupying the following binding sites: phenylbutazone (PBZ; Sudlow's Site I, subdomain IIA), ibuprofen (IBU; Sudlow's Site II, subdomain IIIA), and bilirubin (BIL; Heme binding site, subdomain IB) [9,41]. Site markers (0.5–2.0 μM) were added to samples containing 2 μM ZEN and 2 μM HSA. Because complex formation of ZEN with HSA results in the increased fluorescence signal of ZEN (see in Figs. 1 and 2), displacement of ZEN from HSA leads to the decrease of the fluorescence intensity at 455 nm. IBU had no effect on the spectra of ZEN + HSA solution, while PBZ and BIL decreased the fluorescence signal at 455 nm (Fig. 6A). In order to test the influence of site markers on fluorescence signal of the HSA preparation, measurements were performed in the absence of ZEN as well: PBZ and IBU had negligible effect on the fluorescence intensity at 455 nm, while the fluorescence signal was decreased in the presence of BIL (Fig. 6B). Site markers alone did not give fluorescence signal at 455 nm under the applied conditions and did not influence the fluorescence spectrum of ZEN in the absence of HSA. Furthermore, PBZ, IBU, and BIL at the concentrations applied showed negligible absorbance at the wavelength used for fluorescence measurements reflecting negligible inner filter effect on the fluorescence signal. Since IBU failed to influence the binding of ZEN to HSA, we can exclude the involvement of Sudlow's Site II as the binding site of ZEN. However, the presence of PBZ resulted in decreased fluorescence intensity at 455 nm, therefore subdomains IIA may be involved as a binding site of ZEN on HSA. Although, BIL decreased fluorescence signal even in the absence of ZEN, it is a slight effect compared to the decrease in the presence of ZEN + HSA. Therefore, we cannot exclude subdomain IB as a potential binding site of ZEN. On the other hand, we need to notice that PBZ and BIL resulted in relatively low decrease of the fluorescence signal at 455 nm, despite of both compounds bind higher affinity to HSA compared to ZEN [9,11]. These observations open the possibility that ZEN can occupy another, non-conventional binding site on HSA.

3.6. Involvement of Sudlow's Site I (Subdomain IIA) in ZEN-HSA Interaction

To investigate further the involvement of Sudlow's Site I in ZEN-HSA interaction, the influence of ZEN on albumin-binding of warfarin (WAR) was tested. WAR is the most commonly used site marker of Sudlow's Site I [8]. WAR possesses much stronger fluorescence than ZEN (mainly in the presence of HSA) and because of the spectral interferences we did not apply WAR during our previous investigation (see in Section 3.5). On the other hand, fluorescence signal of ZEN (even in the presence of HSA) is relatively low compared to the albumin-bound WAR, therefore the interaction of WAR with HSA in the presence of ZEN was examined. Complex formation of WAR with HSA results in the remarkable increase of its fluorescence, thus displacement of WAR from HSA accompanied by the significant decrease of its fluorescence signal [23]. In this experiment, 1 μM WAR and 3.5 μM HSA was applied using the fluorescence

excitation and emission wavelength maxima of HSA-bound WAR (317 and 379 nm, respectively). Under these conditions approximately 70% of WAR is presented in albumin-bound form [13]. Contrary to our expectations, in the presence of ZEN (0–10 μM) higher fluorescence intensities of WAR were observed (Fig. 7A). To exclude that the elevation of the fluorescence signal is resulted from spectral interferences, fluorescence spectra of ZEN (0–10 μM) in the absence and presence of WAR (1 μM) or HSA (3.5 μM) were also recorded. The fluorescence emission intensity of ZEN at 379 nm is negligible (even in the presence of HSA) compared to WAR-HSA complex (Fig. 7B). In the absence of HSA, ZEN has no influence on the fluorescence signal of WAR (data not shown). Furthermore, the addition of higher ZEN concentrations (15 and 20 μM) to 1 μM WAR and 3.5 μM HSA did not cause further increase of the fluorescence signal at 379 nm showing that the fluorescence enhancement did not result from the increasing signal of ZEN or ZEN-HSA complex (data not shown). These observations suggest that in the presence of ZEN, the albumin-bound fraction of WAR is increased.

Based on these results, ZEN does not displace WAR from HSA but increases its binding affinity. To confirm this unexpected result, ultrafiltration experiments were performed. Previously, we tried to examine the binding site of ZEN using ultrafiltration technique, however, all tested filters adsorbed the close to total amount of ZEN resulting in its negligible presence in the filtrate. On the other hand, the recovery of WAR after ultrafiltration was relatively high in the filtrate (approximately 80%). For this reason, the effect of ZEN on WAR-HSA interaction can be investigated using ultrafiltration technique. The filtered fraction of WAR (1 μM) was determined in the absence and presence of HSA (5 μM) and/or ZEN (2 and 5 μM). The presence of 5 μM HSA resulted in approximately 50% reduction of WAR in the filtrate, and the simultaneous presence of ZEN in the samples led to its further significant decrease (Fig. 8). In the absence of HSA, ZEN had no influence on the filtered fraction of WAR (data not shown). Because only the free (not albumin-bound) WAR can reach the filtrate, decreased amount of WAR means the reduced fraction of free WAR, suggesting again that the presence of ZEN leads to the increased affinity of WAR towards HSA.

Because contradictory results were observed regarding the two Sudlow's Site I ligands PBZ and WAR, a third molecule was tested as well. The mycotoxin ochratoxin A (OTA) forms very stable complex with HSA, it occupies subdomain IIA showing strong similarities with the albumin binding of WAR [42,43]. Interaction of OTA with HSA can be precisely followed by fluorescence polarization or anisotropy techniques [24]. OTA possesses strong fluorescence property; the binding of OTA to the much larger HSA molecule results in the considerable decrease of its rotational freedom, and therefore the increase of the anisotropy values of OTA. Because the fluorescence wavelength maxima of free and albumin-bound OTA are similar [24], the fluorescence signals of these two forms give a mixed system where increasing displacement of OTA from HSA leads to the gradual decrease of fluorescence

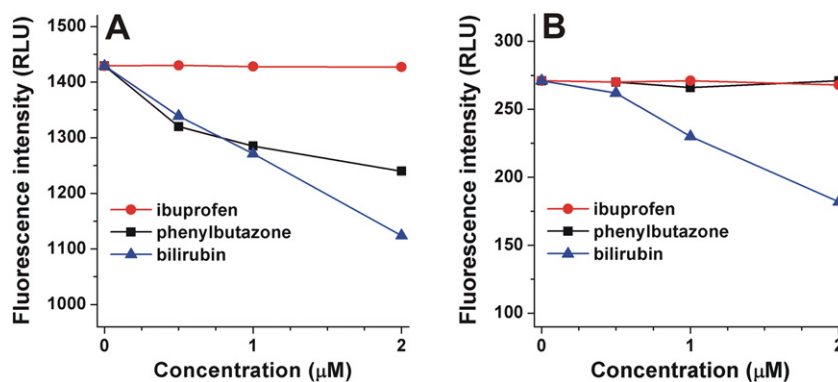


Fig. 6. Effects of site markers (0.5, 1.0, or 2.0 μM) on the fluorescence emission intensities in the presence of ZEN (2 μM) + HSA (2 μM) (A), and in the presence of HSA (2 μM) without ZEN (B) [λ_{exc} = 315 nm, λ_{em} = 455 nm; exc slit width: 10 nm, em slit width: 20 nm].

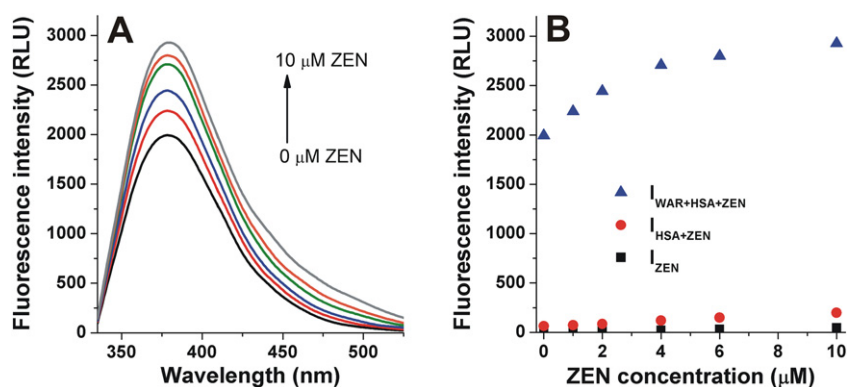


Fig. 7. Fluorescence emission intensities of WAR (1 μM) in the presence of HSA (3.5 μM) and increasing ZEN concentrations (0–10 μM) (A). Fluorescence emission intensities of ZEN (0–10 μM) in the absence and presence of HSA (3.5 μM) or WAR (1 μM) + HSA (3.5 μM) (B) [$\lambda_{\text{exc}} = 317 \text{ nm}$, $\lambda_{\text{em}} = 379 \text{ nm}$; exc slit width: 5 nm, em slit width: 10 nm].

anisotropy. In the presence of 1.0 μM OTA and 1.4 μM HSA, almost the complete amount of OTA is in albumin-bound form [24]. The presence of even 50 μM ZEN caused only negligible effect on the fluorescence anisotropy of OTA, while the presence of WAR strongly decreased anisotropy at the same concentrations (Table S1). We repeated this experiment with 0.75 μM HSA because under these circumstances the albumin-binding of OTA is not complete (approximately 60%) [24]. Neither positive nor negative changes of anisotropy values were observed in the presence of ZEN (data not shown). These results show that ZEN has no influence on the complex formation of OTA with HSA, even in the presence of high ZEN concentrations (15–50 μM).

The results listed above show that PBZ is able to partly displace ZEN, while ZEN does not displace OTA or WAR from HSA. Moreover, the presence of even a few μM ZEN results in the increased binding affinity of WAR towards HSA. These observations suggest that ZEN does not directly interact with Sudlow's Site I but ZEN is able to influence the complex formation of some Site I ligands with HSA, possibly due to the allosteric interaction with the binding site. Furthermore, considering these results and the fact that fluorescence of Trp214 was strongly quenched by ZEN, it is plausible to conclude that the binding site of ZEN is located relatively close to Sudlow's Site I.

3.7. Involvement of Subdomain IB in ZEN-HSA Interaction

To investigate subdomain IB as the potential binding site of ZEN, the displacing ability of ZEN vs. another subdomain IB ligand methyl orange

was tested [11]. The absorption spectrum of methyl orange during its binding to HSA shows a red shift (464 $\rightarrow$ 477 nm) and its absorbance increases in this spectral range (Fig. S2). HSA gives no absorbance at these wavelengths. Methyl orange binds with similar affinity to HSA than ZEN [11]. To test the effect of ZEN on the methyl orange-HSA interaction, increasing ZEN concentrations (0–15 μM) were added to 10 μM methyl orange and 15 μM HSA. ZEN did not influence the absorption spectrum of methyl orange-HSA complex (data not shown), suggesting that the binding site of ZEN is not located on subdomain IB.

Because BIL slightly displaced ZEN from HSA (Fig. 6) as well as the binding site of BIL is allosterically coupled with Sudlow's Site I [8], it is possible that BIL has influence on ZEN-HSA interaction due to the close location of the binding site of ZEN to Sudlow's Site I.

3.8. Testing the Effect of Thyroxine on ZEN-HSA Interaction

Site A (see in Table 1 and Fig. 5) was identified by modeling studies as a potential binding site of ZEN on HSA. Since it is a non-conventional binding site, we have no specific site markers. Thyroxine has more binding site on HSA, and one of these binding sites is located between Sudlow's Site I and II [8] similarly to Site A. Therefore, the influence of thyroxine on the ZEN-HSA interaction was tested. Using the principles described in Section 3.5, increasing concentrations of thyroxine (1–5 μM) were added to 2 μM ZEN and 2 μM HSA. These thyroxine concentrations gave no fluorescence signal under the applied circumstances (PBS, $\lambda_{\text{exc}} = 315 \text{ nm}$, $\lambda_{\text{em}} = 455 \text{ nm}$), and did not influenced the spectra recorded in the presence of ZEN alone (without HSA) or HSA alone (without ZEN). On the other hand, thyroxine led to the concentration dependent decrease of the fluorescence signal at 455 nm when both ZEN and HSA were presented in the solution (Fig. S3). Although, the presence of thyroxine caused relatively slight decrease of fluorescence, we need to consider that many thyroxine molecules occupy other binding sites and do not compete with ZEN, because thyroxine has four other binding sites on HSA [8]. Therefore, the observed effect of thyroxine on the fluorescence of ZEN-HSA complex is in agreement with the hypothesis that the primary binding site of ZEN is located between Sudlow's Site I and Site II.

3.9. Summary of the Investigations Performed to Identify the Binding Site of ZEN on HSA

Based on spectroscopic, ultrafiltration, and molecular modeling studies, we excluded Sudlow's Site I and II as binding sites of ZEN on HSA molecule.

The Heme binding site (on subdomain IB) seems as a possible binding site of ZEN because of the following reasons: docking studies suggested it as a potential high affinity binding site; BIL was able to partly displace ZEN from HSA; and Heme binding site is allosterically coupled

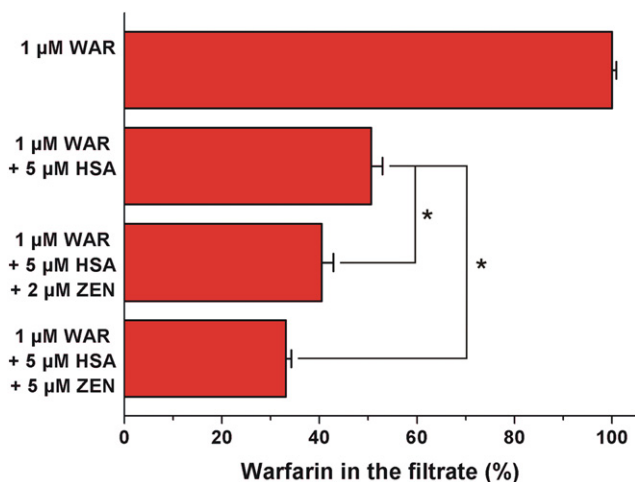


Fig. 8. Occurrence of WAR in the filtrate after ultrafiltration in the absence and presence of HSA (5 μM) and ZEN (2 or 5 μM) [$*p < 0.05$].

with Sudlow's Site I [8]. In contrast, other observations are not readily in line with the involvement of Heme binding site: ZEN was not able to displace methyl orange; BIL was only slightly displaced ZEN from HSA despite its much higher affinity towards albumin; and this binding site is not so close to Trp214. Therefore the involvement of Heme binding site as the high affinity binding site of ZEN is unlikely.

On the other hand, molecular docking studies suggests another high affinity binding site between Sudlow's Site I and II. Our observations are compatible with the view that it is the primary binding site of ZEN on HSA because of the following reasons: this binding site is located close to Sudlow's Site I (containing Trp214 structure) which explains the strong quenching of HSA fluorescence by ZEN; it is in agreement with the results that ZEN can allosterically modulate the interaction of HSA with some Sudlow's Site I ligands; and thyroxine was able to displace ZEN from HSA. Therefore, it is plausible to conclude that the high affinity binding site of ZEN on HSA is Site A (Fig. 5).

4. Conclusions

The interaction of ZEN with HSA was investigated using spectroscopic, ultrafiltration, and molecular modeling techniques. This study proves that ZEN forms complex with HSA, the calculated binding constant suggests a stable and possibly biologically relevant interaction. Our results recommend that ZEN has one high affinity binding site on HSA which is located between Sudlow's Site I and II. It is a non-conventional binding site, however, because of its close location to Sudlow's Site I, ZEN is able to influence the complex formation of some Sudlow's Site I ligands with HSA. Similarly, some Sudlow's Site I ligands (e.g., phenylbutazone) can affect the ZEN-HSA interaction. Further in vivo experiments are required in order to explore the biological importance of ZEN-albumin complex formation as well as to investigate the influence of different drugs and xenobiotics on albumin binding of ZEN.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jphotobiol.2017.03.016>.

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